AMAL JYOTHI COLLEGE OF ENGINEERING

Kanjirappally, Kerala

DATA STRUCTURES AND ALGORITHM MINI PROJECT REPORT

SMART CHARGING STATION MANAGER

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1. ABSTRACT

The Smart Charging Station Manager is a JavaScript-based browser application designed to organize electric vehicle (EV) charging slots and waiting vehicles. As EV usage increases, charging stations need a simple way to manage vehicle arrivals, slot availability, and charging information. This project uses arrays to represent charging slots and vehicle records, and a queue to manage waiting vehicles using the First-In-First-Out (FIFO) principle. When a slot becomes free, the vehicle at the front of the queue is automatically allocated to it. The application also supports vehicle registration, searching, charging-session completion, estimated charging time and cost, and browser-based data storage. The project demonstrates the practical use of data structures and basic algorithms in a user-friendly interface.

2. INTRODUCTION

Electric vehicles are becoming an important part of sustainable transportation. A charging station may need to handle several vehicles and charging slots at the same time. Without an organized system, it can be difficult to identify available slots and maintain a fair waiting order. Smart Charging Station Manager provides a simple digital dashboard for registering vehicles, monitoring charging sessions, and managing a waiting queue.

3. PROBLEM STATEMENT

Manual tracking of charging slots can cause confusion about slot availability and the order in which waiting vehicles should be served. A simple software solution is needed to store vehicle information, identify free slots, and allocate a newly available slot to the next waiting vehicle.

4. OBJECTIVES

• To manage a fixed set of EV charging slots. • To register and store vehicle details. • To implement a FIFO queue for waiting vehicles. • To search registered vehicles. • To calculate approximate charging time and cost. • To provide a responsive and easy-to-use interface.

5. TECHNOLOGIES USED

HTML5 is used to structure the application, CSS3 provides the visual design and responsive layout, and vanilla JavaScript implements application logic and data structures. Browser Local Storage preserves the demonstration data across page refreshes. The project runs locally in a modern web browser and does not require a backend server.

6. DATA STRUCTURES AND ALGORITHMS

Arrays: An array represents the six charging slots and stores registered vehicle records. Each slot stores the ID of the vehicle currently using it, if any. Queue (FIFO): A queue stores IDs of vehicles waiting for a slot. New arrivals are appended to the end, and allocation removes the vehicle at the front. Searching: The vehicle manager filters records by registration number, status, or slot. Slot allocation: The application scans for the first available slot. Estimation algorithms: Remaining energy is estimated from battery capacity and current battery percentage; charging time is estimated using the selected charger rate, and cost uses an illustrative tariff of ■8.50 per kWh.

7. SYSTEM DESIGN AND WORKING

The user opens the dashboard and registers a vehicle with its registration number, battery level, battery capacity, charging rate, and priority. If a slot is free, the vehicle is allocated immediately. If all slots are occupied, its ID is appended to the FIFO queue. When the user marks a charging session complete, the slot is freed and the first vehicle in the queue is allocated automatically. Vehicle records can be searched, and data is saved in browser Local Storage.

8. FEATURES

• Dashboard with total, available, occupied, and waiting counts. • Six visual charging-slot cards. • Vehicle registration form. • Automatic slot allocation. • FIFO waiting queue. • Searchable vehicle records. • Manual completion of charging sessions. • Estimated time and cost. • Recent activity list. • Responsive design and Local

Storage persistence.

9. OUTPUT

The dashboard displays six charging slots and sample vehicles when first opened. Four slots are initially occupied and two sample vehicles are waiting, allowing the FIFO behavior to be demonstrated immediately. The vehicle manager allows new registrations and searching. Completing a charging session releases its slot and automatically moves the next waiting vehicle into that slot. The included output_preview.png is an illustrative dashboard preview; the live output is generated by opening index.html in a browser.

10. TESTING

Test 1: Open index.html and verify the dashboard and sample records appear. Test 2: Register a new vehicle while a slot is free and verify automatic allocation. Test 3: Fill all six slots and register another vehicle; verify that it joins the queue. Test 4: Complete one charging session and verify that the first waiting vehicle receives the free slot. Test 5: Search by registration number and verify that matching records appear. Test 6: Refresh the browser and verify that the state is retained. Test 7: Use Reset to restore the initial demonstration data.

11. LIMITATIONS

This is an academic prototype and does not communicate with physical charging hardware. Charging completion is manually triggered. Charging time is approximate and does not model real battery charging curves or power losses. The ■8.50/kWh tariff is illustrative, not a live electricity rate. Local Storage data belongs to the current browser profile and is not shared between users. The priority field is recorded, but allocation remains FIFO so the queue behavior stays consistent with the project's data-structure objective.

12. FUTURE ENHANCEMENTS

Future versions could include a database, secure user authentication, real charger telemetry, automatic charging-session updates, configurable tariffs, downloadable reports, and a more advanced scheduling policy.

13. CONCLUSION

Smart Charging Station Manager demonstrates how arrays, queues, searching, and basic algorithms can solve a practical resource-management problem. It organizes charging slots, maintains a FIFO waiting list, and provides a clear dashboard for vehicle management. The project combines core Data Structures concepts with a responsive web interface and can be extended into a more complete charging-station application.

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